…… Massive Crowd Management in Urban Pedestrian Zones

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***Abstract*—This paper presents a bi-level optimization framework for crowd management in large pedestrian zones, integrating discrete directional constraints with anisotropic geometric guidance. The lower level employs a Hughes-type macroscopic crowd model enhanced with constrained Hamilton-Jacobi-Bellman (HJB) equations for one-way, bidirectional, and closed channel states, and then uses metric tensor fields to represent soft channel-alignment guidance within the feasible direction set. The upper level optimizes channel direction configurations and guidance intensities using a Structure-Aware Hybrid Block Optimization (SA-HBO) algorithm. Three raw physical metrics are introduced for efficiency, safety, and load balancing, and are further converted into dimensionless indicators for cross-scheme comparison and scalarization. The proposed method exploits the local response characteristics of Bellman potential fields and employs surrogate pre-screening with Bellman warm-start strategies to reduce computational cost. Numerical experiments demonstrate the effectiveness of the proposed approach.**

***Index Terms*—Crowd dynamics, macroscopic modeling, channel optimization, Hamilton-Jacobi-Bellman equation, bi-level optimization.**

I. Introduction

LARGE-SCALE crowd gathering risks not only threaten individual safety but may also escalate into stampede incidents that disrupt social order and public security; the effectiveness of crowd management directly tests urban governance capacity and the modernization level of social safety governance systems [1], [2]. With the increasing frequency of high-density pedestrian activities such as holiday tourism, urban public events, commercial district promotions, New Year celebrations, and major sports events, core urban areas must accommodate pedestrian demands far exceeding daily levels within short time windows, posing enormous challenges for on-site organization, passage management, risk early warning, and emergency dispatch [3], [4], [5]. Against this backdrop, scientific modeling, numerical simulation, and optimization methods for formulating and evaluating crowd control strategies—improving the efficiency of large-scale event organization and reducing local congestion risks—have become critical scientific problems demanding urgent breakthroughs in social safety and urban governance [6], [7].

Crowds in large-scale events constitute typical complex systems. Numerous individuals with heterogeneous desired speeds, destinations, and route preferences interact through multi-scale, nonlinear mechanisms in confined spaces, generating spatiotemporal patterns at the collective level [8], [4]. Macroscopically, these patterns manifest as destination-oriented flow, passage convergence, bottleneck queuing, exit competition, and directional responses to control regulations [9], [10]; microscopically, individuals are influenced by surrounding density, visible paths, obstacle boundaries, and guidance facilities [11]. For open scenic areas, crowd behavior additionally exhibits pronounced phase dependence: visitors may first move toward the viewing platform, then tour along the main viewing direction, and subsequently choose different exit passages based on preferences [12]. These characteristics of multi-stage, multi-route coexistence with shared congestion mean that traditional static capacity analysis or single shortest-path models cannot adequately support refined management decisions.

Among various urban crowd gathering scenarios, open scenic areas, waterfront viewing platforms, and historic commercial districts are highly representative. Unlike enclosed venues such as stadiums and subway stations with well-defined boundaries [13], [14], such areas consist of open spaces, main tourist corridors, multiple lateral connecting passages, and return flow lines; visitors typically undergo a continuous behavioral chain of "entry–touring along the main corridor–choosing passages to leave–returning" [15], with crowd movement governed jointly by spatial geometry, control regulations, and historical behavioral preferences [16], [15]. Taking the Shanghai Nanjing East Road–Bund area as a concrete example, visitors approach the Bund from Nanjing East Road, ascend to the viewing platform via multiple stepped passages, tour along the waterfront, then exit through several southern passages and return along lower streets. This scenario can be abstracted as the typical "waterfront platform–multi-stepped passage" crowd flow scenario. In such scenarios, management difficulties manifest in three aspects: first, multiple passages simultaneously serve entry, exit, and diversion functions, and local direction settings can alter overall flow organization; second, distinct phase-dependent behavioral transitions exist between the main tourist corridor and lateral passages, making single origin–destination models insufficient; third, local congestion and global route choice form mutual feedback loops, where control changes in one passage may trigger load redistribution in adjacent areas. Fig. 1 shows the large-scale crowd at the Shanghai Nanjing East Road–Bund scenario during a holiday period.

Existing large-scale event security management practices typically integrate system dynamics, system coupling, and system safety theory, combined with high-risk zone delineation and historical crowd behavior patterns, to formulate a series of control measures [17], [18]. These measures include deploying security personnel at critical locations, erecting temporary barriers for isolation and diversion, implementing one-way passage on key channels, and applying batch release in core areas [19], [20]. For instance, during major holidays, the Shanghai Bund area adopts batch release, passage direction regulation, south-entry north-exit routing, and reduction of

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Fig. 1. Large-scale crowd at the Shanghai Nanjing East Road–Bund scenario during a holiday period.

head-on conflicts [21]; similarly, Nanluoguxiang main street in Beijing implements one-way passage measures (south entry, north exit) during peak periods [22], consistent in control mechanism with the Shanghai Bund approach.

However, current management measures for urban high-pedestrian-flow areas still rely considerably on manual experience and static contingency plans, with deficiencies remaining in scientific evaluation and refined optimization [23]. First, many control schemes are formulated based on historical experience, lacking quantitative characterization of the impact of different passage direction configurations, geometric guidance intensities, and phase-dependent behavioral preferences. Second, existing methods tend to focus on single-point statistics or local density monitoring, making it difficult to describe the dynamic coupling among "passage direction rules–local optimal directions–collective density evolution–exit load distribution" [24]. Third, in open scenic area scenarios with multiple entrances, passages, and stages, neglecting visitor route preferences and phase transition behaviors can easily lead to misjudgment of passage loads and high-risk area locations. Finally, many existing simulation or optimization methods treat the crowd model as a black-box evaluator, failing to exploit the structural information among passage geometry, one-way rules, and multi-route subpopulations [25], resulting in insufficient efficiency in control strategy search [26].

To address the above issues, this paper proposes a macroscopic crowd modeling and control optimization method incorporating passage constraints, geometric guidance, and multi-stage route behavior, targeting the "waterfront platform–multi-stepped passage" open scenic area scenario. At the modeling level, this paper constructs a Bellman–conservation law coupled crowd continuum model: different stage–route subpopulations possess independent potential functions with optimal directions computed via the discrete Bellman equation, while sharing the speed–density relationship determined by total density; one-way, bidirectional, and closed passage states are formulated as constraints on the local admissible direction set, and an anisotropic metric tensor characterizes the cost differences between axial and lateral movement, simulating pedestrian pre-alignment and streamline reorganization near passages. At the optimization level, this paper distinguishes "controllable management variables" (passage direction configuration and geometric guidance intensity) from "exogenous behavioral parameters" (visitor route diversion preferences identified from historical data). Based on this boundary, total travel time, high-density exposure time, and channel cumulative flow variance are adopted as efficiency, safety, and load balancing indicators, forming a scalarized optimization problem under fixed weights. A simplified instance of the Structure-Aware Hybrid Block Optimization framework (SA-HBO) is designed to solve this problem, exploiting the discrete neighborhood structure of passage direction variables, the continuous nature of geometric guidance variables, and the response characteristics of Bellman potential fields to local control changes, improving candidate strategy evaluation efficiency through surrogate pre-screening, full simulation confirmation, and Bellman warm-start procedures.

The main contributions of this paper are as follows.

1) A "waterfront platform–multi-stepped passage" crowd flow modeling framework for open scenic areas is proposed, representing visitor behavior as a multi-stage, multi-route continuum flow process, where different subpopulations possess independent potential functions while sharing collective congestion feedback.
2) A Bellman–conservation law coupled model integrating passage constraints and geometric guidance is constructed. The admissible direction set first encodes one-way, bidirectional, and closed channel states, while an anisotropic metric tensor simulates pre-alignment, smooth convergence, and streamline reorganization near passage entrances.
3) An evaluation framework distinguishing mechanism verification from strategy optimization is established, categorizing indicators into behavioral mechanism indicators (direction consistency, approach angle distribution, channel capture domain) and management optimization indicators (total travel time, high-density exposure time, channel flow variance).
4) A structure-aware optimization framework for controllable passage configuration is designed, jointly optimizing passage direction configuration and geometric guidance intensity under fixed historically identified preference parameters through discrete neighborhood search, continuous local search, and surrogate pre-screening with acceptance/rejection mechanisms.

II. Related Work

A. Massive Crowd Management and Control

Crowd management in urban centers has been extensively studied from control-theoretic and optimization perspectives, aiming to enhance safety and efficiency through boundary regulation, facility layout, and intelligent decision-making [27], [28].

In control-theoretic approaches, Zhong et al. [29] designed boundary control strategies for urban traffic flow networks to ensure convergence to uncongested states under disturbances, with stability analysis based on Lyapunov methods [30]. For one-dimensional crowd dynamics, Wadoo and Kachroo [31] employed advection and diffusion control to prevent blocking and shock waves during evacuation. Addressing multi-directional disturbance propagation, Qin [32] proposed a unit sliding mode controller based on integral barrier Lyapunov functions to ensure boundedness of internal state variables, demonstrating strong disturbance rejection capabilities. For heterogeneous corridor regulation, Zhu et al. [33] combined policy learning with neural networks to learn optimal feedback controllers that adjust inflow rates and free-flow speeds, achieving balanced density distribution and reduced congestion.

Physical infrastructure regulation has also proven effective. Studies have shown that strategically placed obstacles can significantly improve outflow, velocity, and local density in merging areas [23], [34]. Yang et al. [35], [36] employed Model Predictive Control (MPC) to optimize barrier lengths at subway bottlenecks, later extending control variables to entrance limiters, gates, and escalator directions for dynamic, real-time crowd regulation. For evacuation guidance, Carmona and Paricio-Garcia [37] proposed dynamic exit-choice recommendations using multinomial Logit models. However, their approach relies on discrete speed instructions (e.g., "stop," "walk," "run") transmitted via electronic wristbands [38], which poses implementation challenges in open urban scenarios.

Recently, intelligent optimization and deep reinforcement learning have emerged as promising alternatives. Liao et al. [39] formulated fence layout as a subset selection problem, while subsequent work [26] proposed a dynamic crowd inflow control system using LSTM networks to capture temporal features and Proximal Policy Optimization (PPO) [40] for real-time entrance flow regulation. Differential Evolution (DE) [41] has also been applied to optimize guardrail layouts and flow guidance in real-world scenarios such as Chengdu East Railway Station, using average travel time and crowd pressure as multi-objective criteria [42].

B. Macroscopic Crowd Modeling

Macroscopic crowd models treat pedestrian flows as continuous media, offering computational efficiency and analytical tractability compared to microscopic approaches. Henderson [43] first applied fluid dynamics to pedestrian traffic in the early 1970s, comparing crowd motion to the behavior of gas or fluid particles.

Hughes [44] pioneered the first-order continuum theory for pedestrian flow, introducing the classical model in which pedestrian walking speed is governed by local density through a fundamental diagram, while movement direction follows the steepest-descent path of a potential function representing the remaining travel time to exits. This framework elegantly captures the interplay between congestion and route choice. Ling et al. [45] subsequently generalized the Hughes formulation and developed an efficient numerical solution combining a weighted essentially non-oscillatory (WENO) scheme for the conservation law with a fast sweeping method for the eikonal equation.

Several extensions have enriched the descriptive capability of macroscopic models. Nonlocal flux models were proposed by Colombo et al. [46], replacing pointwise density with a neighborhood-averaged quantity in the velocity function to account for anticipatory behavior. This direction was further extended to incorporate anisotropic perception fields and boundary effects from walls and obstacles, with high-resolution numerical schemes and well-posedness results suitable for realistic architectural configurations [47]. To enforce the hard incompressibility constraint—that pedestrian density cannot exceed a physically admissible maximum—Maury et al. [48] introduced the projection model, which formulates the actual velocity as a least-squares projection of the desired velocity onto the set of feasible (non-overlapping) configurations. This framework was later recast as a Wasserstein gradient flow with density constraints [49], using the Jordan–Kinderlehrer–Otto (JKO) scheme [50] to discretize the motion while guaranteeing that density respects an upper bound, making it well-suited for high-density congestion scenarios.

To overcome non-physical behaviors such as supersonic information propagation that may arise in first-order models, Aw and Rascle [51] introduced a second-order model based on a generalized momentum conservation law that preserves anisotropy. The Aw–Rascle (AR) framework was subsequently adapted to pedestrian flows [52], where the congestion pressure term successfully reproduces characteristic phenomena including stop-and-go waves and spontaneous lane formation.

Beyond the classical Hughes-type framework, extensions have incorporated control and guidance mechanisms relevant to crowd management. Anisotropic formulations using metric tensors have been developed to represent channel geometries and directional preferences [53], [54], allowing for continuous geometric guidance without explicit prohibition. Constrained Hamilton-Jacobi-Bellman (HJB) formulations have been employed to model strict directional constraints such as one-way corridors, with the Bellman optimality principle providing a natural framework for unifying anisotropic geometric guidance with discrete directional constraints.

Despite these advances, existing approaches often treat control parameters as fixed or manually tuned. The bi-level optimization of discrete channel direction configurations combined with continuous guidance intensities remains underexplored, particularly for large-scale urban pedestrian zones with multiple behavioral stages and heterogeneous route choices.

III. Macroscopic Crowd Dynamics Model

A. Hughes Model Foundation

We consider a bounded two-dimensional domain $\Omega \subset \mathbb{R}^2$ representing the pedestrian walkway. Let $\rho(x,t)$ denote the crowd density at position x and time t, $\phi(x,t)$ the minimal remaining time to reach an exit, and $\mathbf{v}(x,t)$ the mean velocity field. The continuity equation governing mass conservation reads:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0. \tag{1}$$

In the classical Hughes model [44], pedestrians move in the direction of steepest descent of the potential function:

$$\mathbf{v} = f(\rho)\frac{-\nabla\phi}{|\nabla\phi|}, \tag{2}$$

where $f(\rho)$ is the speed-density relationship. We adopt the Greenshields form [55]:

$$f(\rho) = v_{\max}\left(1 - \frac{\rho}{\rho_{\max}}\right), \tag{3}$$

with $v_{\max}$ the free-flow speed and $\rho_{\max}$ the jam density.

The potential function satisfies the eikonal equation:

$$|\nabla\phi| = \frac{1}{f(\rho)}. \tag{4}$$

While this model captures basic congestion effects, it assumes that all directions are feasible and equally weighted. It therefore cannot represent one-way rules, temporary closures, or channel-induced geometric preferences. We address these limitations by first introducing HJB admissible-direction constraints for hard channel rules and then introducing the metric tensor to encode soft geometric guidance within the feasible direction set.

B. Channel Control via Constrained HJB

Many crowd management measures directly restrict available movement directions rather than merely adjusting traversal costs. Examples include one-way corridors, bidirectional passages, and temporary closures. Such rules are hard feasibility constraints on the local action set and are naturally modeled via constrained optimal control.

Let $U(x)$ denote the set of admissible unit directions at position x. The velocity satisfies:

$$\mathbf{v} = f(\rho)\mathbf{u}, \quad \mathbf{u}(x) \in U(x), \quad |\mathbf{u}| = 1. \tag{5}$$

By the Bellman optimality principle, the potential function satisfies the constrained HJB equation:

$$\max_{\mathbf{u}\in U(x)} \{-f(\rho)\mathbf{u}\cdot\nabla\phi\} = 1. \tag{6}$$

When $U(x)$ is the full unit circle, (6) reduces to the classical Hughes model. For restricted control sets, various channel rules can be expressed naturally. For channel c with region Ω_c and reference tangent $\tau_c(x)$, define

$$s_c \in \{-1, 0, +1, \varnothing\}, \tag{7}$$

where $+1$, -1, 0, and $\varnothing$ denote forward one-way, reverse one-way, bidirectional, and closed states, respectively. The corresponding admissible set is

$$U_c(x; s_c) = \begin{cases} \{\tau_c(x)\}, & s_c = +1, \\ \{-\tau_c(x)\}, & s_c = -1, \\ \{\tau_c(x), -\tau_c(x)\}, & s_c = 0, \\ \varnothing, & s_c = \varnothing, \end{cases} \quad x \in \Omega_c. \tag{8}$$

For a closed channel, the continuous maximum over an empty set is not used as an ordinary HJB state. Numerically, this state is handled as no feasible control in the channel region, equivalently $\phi(x) = +\infty$ for unreachable cells or infinite Bellman step cost from those cells.

C. Channel Geometric Guidance via Metric Tensor

The admissible set $U(x)$ determines which directions are feasible, but it does not fully describe how pedestrians approach and align with channel geometry before entering a passage. In real scenes, pedestrians often pre-align with the channel axis and merge smoothly near entrances rather than abruptly changing direction at the channel boundary. Therefore, after defining hard HJB constraints, we introduce a metric tensor $\mathbf{M}(x)$ to encode soft geometric preference within the feasible direction set.

To represent anisotropic channel geometries such as corridors and barriers, we introduce a spatially varying symmetric positive-definite metric tensor $\mathbf{M}(x)$ that generalizes the classical eikonal equation to anisotropic form [53]:

$$\sqrt{\nabla\phi^\top \mathbf{M}(x)\nabla\phi} = \frac{1}{f(\rho)}. \tag{9}$$

Under this metric, the optimal direction is determined by the tensor-induced steepest descent:

$$\mathbf{v} = f(\rho)\frac{-\mathbf{M}(x)\nabla\phi}{\sqrt{\nabla\phi^\top \mathbf{M}(x)\nabla\phi}}. \tag{10}$$

For a channel region Ω_c with unit tangent $\tau(x)$ and unit normal $n(x)$, we construct the metric tensor as:

$$\mathbf{M}(x) = \alpha(x)\tau\tau^\top + \beta(x)nn^\top, \quad x \in \Omega_c, \tag{11}$$

where $\alpha(x)$ and $\beta(x)$ control the traversal weights in tangential and normal directions, respectively. In the discrete Bellman update used in this paper, the directional step cost is proportional to $1/\sqrt{\mathbf{u}^\top\mathbf{M}(x)\mathbf{u}}$. Hence setting $\alpha(x) \gg \beta(x)$ lowers the effective cost along the channel tangent and suppresses cross-channel deviations, achieving continuous geometric guidance without explicit prohibition. Outside channels, $\mathbf{M}(x) = \mathbf{I}$ recovers the isotropic case.

Remark: The metric tensor represents directional preference rather than hard feasibility. As long as costs remain finite, pedestrians may still deviate when congestion or detours require it. Thus $U(x)$ answers which directions are allowed, whereas $\mathbf{M}(x)$ answers which allowed directions better follow the channel geometry.

D. Unified Discrete Bellman Formulation

To unify anisotropic geometry with discrete constraints, we employ a discrete Bellman formulation. For grid point x and admissible direction $\mathbf{u} \in U(x)$, define the step cost [56]:

$$\mathrm{Cost}(x, \mathbf{u}) = \frac{\Delta x}{f(\rho(x))}\frac{1}{\sqrt{\mathbf{u}^\top\mathbf{M}(x)\mathbf{u}}}, \tag{12}$$

The potential satisfies:

$$\phi(x) = \min_{\mathbf{u}\in U(x)} \left[\phi(x + \Delta x\,\mathbf{u}) + \mathrm{Cost}(x, \mathbf{u})\right]. \tag{13}$$

This formulation couples metric tensor $\mathbf{M}(x)$ for geometric guidance with control set $U(x)$ for directional constraints within a unified Bellman framework. The optimal direction is:

$$\mathbf{u}^*(x) = \arg\min_{\mathbf{u}\in U(x)} \left[\phi(x + \Delta x\,\mathbf{u}) + \mathrm{Cost}(x, \mathbf{u})\right], \tag{14}$$

with velocity $\mathbf{v}(x) = f(\rho)\mathbf{u}^*(x)$.

E. Multi-Stage Multi-Route Model

Real-world scenarios often involve complex itineraries with multiple stages and route choices. We extend the density field to include stage and route indices, with fixed-probability splitting at decision boundaries.

1) Stage-Route Decomposition: Let the complete behavior chain consist of S stages, where stage s has R_s alternative routes. The total density decomposes as [57]:

$$\rho(x,t) = \sum_{s=1}^{S} \sum_{r=1}^{R_s} \rho_{s,r}(x,t), \tag{15}$$

where $\rho_{s,r}(x,t)$ is the density of sub-population in stage s following route r.

All sub-populations share the same speed function $f(\rho)$ determined by total density, ensuring collective congestion effects. Sub-populations differ in their potential functions, optimal directions, and exit targets.

2) Sub-Population Potentials and Velocities: For each sub-population (s,r), define admissible direction set $U_{s,r}(x)$, metric tensor $\mathbf{M}_{s,r}(x)$, and target regions. The potential $\phi_{s,r}(x)$ satisfies:

$$\begin{aligned}\phi_{s,r}(x) = \min_{\mathbf{u}\in U_{s,r}(x)} \Big[& \phi_{s,r}(x+\Delta x\,\mathbf{u}) \\ & + \frac{\Delta x}{f(\rho)} \frac{1}{\sqrt{\mathbf{u}^\top \mathbf{M}_{s,r}(x)\mathbf{u}}} \Big].\end{aligned} \tag{16}$$

The optimal direction is:

$$\begin{aligned}\mathbf{u}^*_{s,r}(x) = \arg\min_{\mathbf{u}\in U_{s,r}(x)} \Big[& \phi_{s,r}(x+\Delta x\,\mathbf{u}) \\ & + \frac{\Delta x}{f(\rho)} \frac{1}{\sqrt{\mathbf{u}^\top \mathbf{M}_{s,r}(x)\mathbf{u}}} \Big],\end{aligned} \tag{17}$$

with velocity $\mathbf{v}_{s,r}(x) = f(\rho)\,\mathbf{u}^*_{s,r}(x)$.

3) Stage Transitions and Fixed-Probability Splitting: Sub-populations couple through stage transitions. Let $G_{s,r} \subset \Omega$ be the completion goal region for stage (s,r) with indicator $\chi_{G_{s,r}}(x)$. Upon reaching this region, pedestrians switch to next-stage routes with fixed probabilities.

Let $p_{(s,r)\to q}$ be the fixed probability of switching from (s,r) to $(s+1,q)$, satisfying:

$$p_{(s,r)\to q} \geq 0, \quad \sum_{q=1}^{R_{s+1}} p_{(s,r)\to q} = 1. \tag{18}$$

The transition term is:

$$Q_{(s,r)\to(s+1,q)}(x,t) = p_{(s,r)\to q}\,\kappa_{s,r}\,\chi_{G_{s,r}}(x)\,\rho_{s,r}(x,t), \tag{19}$$

where $\kappa_{s,r} > 0$ is the stage switching rate.

The density evolution for each sub-population includes advection and transition source terms:

$$\frac{\partial \rho_{s,r}}{\partial t} + \nabla\cdot(\rho_{s,r}\mathbf{v}_{s,r}) = Q^{\text{in}}_{s,r} - Q^{\text{out}}_{s,r}, \tag{20}$$

where $Q^{\text{in}}_{s,r}$ is inflow from the previous stage and $Q^{\text{out}}_{s,r}$ is outflow to the next stage.

IV. Numerical Solution

We discretize the domain using a uniform two-dimensional grid, with the control direction set represented as an 8-neighborhood basis with masking for regional restrictions.

Unified Bellman Solver. Rather than separate eikonal and HJB solvers, we use a unified discrete Bellman shortest-path update. For candidate direction $\mathbf{u}$, precompute geometric step factors from the local tensor, then combine with the speed function to form the complete step cost. A priority queue propagates potential values outward from exits until convergence. When the control set is unrestricted, this procedure is equivalent to monotone solution of the anisotropic eikonal equation; when controls are restricted, it naturally reduces to constrained HJB solution, covering all required scenarios within a unified framework.

Density Update. After obtaining converged potentials, recover velocity fields using (17), substitute into the continuity equation, and advance density explicitly using an upwind flux scheme:

$$\rho^{n+1} = \rho^n - \Delta t\,\nabla\cdot(\rho^n \mathbf{v}^n). \tag{21}$$

The time step satisfies the CFL stability condition.

Evaluation Oracle. For any given control configuration $z = (s, \eta)$, the simulation workflow constitutes a lower-level evaluation operator:

$$(\rho, \phi, \mathbf{v}) = \mathcal{S}(z; \hat{p}), \tag{22}$$

where $\hat{p}$ represents fixed route preference parameters identified from historical data. This operator provides the foundation for subsequent control optimization.

V. Control Evaluation Metrics

We distinguish between behavior validation metrics (for verifying model mechanisms) and control evaluation metrics (for assessing policy impacts). This section focuses on the latter.

A. Efficiency Metric: Total Travel Time

For simulation horizon $[0,T]$, the total travel time metric is:

$$J_1(z \mid \hat{p}) = \int_0^T \int_\Omega \rho(x,t)\,dx\,dt. \tag{23}$$

This measures aggregate dwell time of the crowd, providing a stable basis for comparing different control strategies.

B. Safety Metric: High-Density Exposure Time

Let ρ_{safe} be the safe density threshold. The exposure time metric is:

$$J_2(z \mid \hat{p}) = \int_0^T \int_\Omega \mathbf{1}_{\rho(x,t) > \rho_{\text{safe}}}\,dx\,dt. \tag{24}$$

This captures cumulative space-time exposure to dangerous crowding conditions, reflecting management concerns about sustained congestion risks.

C. Balance Metric: Channel Flow Variance

Let $q_c(t)$ be the flux through channel c at time t, with cumulative flow $F_c = \int_0^T q_c(t)\,dt$. The load balancing metric is:

$$J_3(z \mid \hat{p}) = \mathrm{Var}(F_1, \ldots, F_m). \tag{25}$$

Lower variance indicates more even distribution of traffic across channels, reducing risk concentration and improving facility utilization.

D. Dimensionless Normalization

Because J_1, J_2, and J_3 have different physical units and numerical scales, we further define normalized indicators for policy comparison and optimization. Let $M_{\text{tot}} = M_{\text{init}} + M_{\text{in}}$ denote the total evaluated crowd mass, let $|\Omega_w|$ be the walkable area, and let $F = \sum_{c=1}^{m} F_c$ be the total cumulative channel flow. We define

$$\tilde{J}_1(z \mid \hat{p}) = \frac{J_1(z \mid \hat{p})}{M_{\text{tot}} T}, \tag{26}$$

$$\tilde{J}_2(z \mid \hat{p}) = \frac{J_2(z \mid \hat{p})}{|\Omega_w| T}, \tag{27}$$

and

$$\tilde{J}_3(z \mid \hat{p}) = \frac{\mathrm{Var}(F_1, \ldots, F_m)}{F^2(m-1)/m^2} = \frac{m^2}{m-1}\mathrm{Var}(p_1, \ldots, p_m), \tag{28}$$

where $p_c = F_c/F$. The normalized metrics are dimensionless, with $\tilde{J}_2 \in [0,1]$ and $\tilde{J}_3 \in [0,1]$.

E. Scalarized Optimization Objective

Given weights $(\lambda_1, \lambda_2, \lambda_3)$, the scalarized objective is:

$$J(z \mid \hat{p}) = \lambda_1 \tilde{J}_1(z \mid \hat{p}) + \lambda_2 \tilde{J}_2(z \mid \hat{p}) + \lambda_3 \tilde{J}_3(z \mid \hat{p}), \tag{29}$$

with the optimization problem:

$$\min_{z \in \mathcal{Z}} J(z \mid \hat{p}). \tag{30}$$

Our algorithm solves this scalarized problem for fixed weights, rather than directly generating the full Pareto frontier. Approximate Pareto sets can be obtained by running with different weight combinations.

VI. Optimization Method

A. Control Variable Definition

We distinguish controllable variables from exogenous behavioral parameters:

Controllable variables:

1) *Channel direction configuration* $s = (s_1, \ldots, s_C)$, where $s_c \in \{-1, 0, +1, \varnothing\}$ indicates reverse one-way, bidirectional, forward one-way, or closed state for channel c.
2) *Geometric guidance intensity* $\eta = (\eta_1, \ldots, \eta_C)$, where $\eta_c = \alpha_c/\beta_c \geq 1$ is the anisotropy ratio.

Thus the optimization variable is $z = (s, \eta)$.

Exogenous parameters: Route preference parameters $\hat{p}$ are identified from historical data and fixed during optimization. Decision region locations are reserved for future extension.

B. Structure-Aware Hybrid Block Optimization

Treating this as a black-box optimization ignores three key structures: mixed discrete-continuous variables, local Bellman response characteristics, and the expensive multi-group simulation structure.

We propose Structure-Aware Hybrid Block Optimization (SA-HBO), with four core components:

1) Discrete block neighborhood search for channel directions;
2) Surrogate pre-screening to reduce full simulations;
3) Bellman warm-start with local re-solving for efficient evaluation;
4) Continuous block projected stochastic approximation for guidance intensities.

C. Discrete Block Update

Fixing continuous variables η^k, the discrete subproblem at iteration k is:

$$s^{k+1/2} \approx \arg\min_{s \in \mathcal{S}} J(s, \eta^k \mid \hat{p}). \tag{31}$$

Define the neighborhood of current configuration s^k as:

$$\mathcal{N}(s^k) = \left\{ s : \|s - s^k\|_0 \leq \nu \right\}, \tag{32}$$

where ν is the maximum number of simultaneously flipped channels (typically 1 or 2).

Surrogate pre-screening: To reduce full simulations, construct a local surrogate for each candidate $s' \in \mathcal{N}(s^k)$:

$$\widehat{J}(s', \eta^k) = \mu_1 \widehat{J}_1^{\text{local}} + \mu_2 \widehat{J}_2^{\text{local}} + \mu_3 \widehat{J}_3^{\text{flow}}, \tag{33}$$

where the local metrics approximate density changes, exposure changes, and flow redistribution. Select top K candidates by surrogate value for full PDE simulation.

D. Continuous Block Update

Fixing discrete variables $s^{k+1/2}$, solve for η:

$$\eta^{k+1} \approx \arg\min_{\eta \in \mathcal{H}} J(s^{k+1/2}, \eta \mid \hat{p}), \tag{34}$$

with feasible set $\mathcal{H} = \{\eta : \eta_c \geq 1\}$.

Since gradients are unavailable analytically, we use projected stochastic approximation. Generate Rademacher perturbation $\Delta_\eta^k \in \{-1, +1\}^C$, construct two-sided perturbations $\eta_\pm^k = \eta^k \pm c_k \Delta_\eta^k$, and project to feasible set. The stochastic gradient estimate is:

$$\widehat{g}_\eta^k = \frac{J(s^{k+1/2}, \tilde{\eta}_+^k \mid \hat{p}) - J(s^{k+1/2}, \tilde{\eta}_-^k \mid \hat{p})}{2c_k} \odot (\Delta_\eta^k)^{-1}. \tag{35}$$

The candidate update is:

$$\eta_{\text{cand}}^{k+1} = \Pi_{\mathcal{H}}\left(\eta^k - \alpha_k D_\eta^{-1} \widehat{g}_\eta^k\right), \tag{36}$$

with diagonal preconditioner D_η.

Acceptance/Rejection: Compare candidate objective J_{cand} with current J_{cur}. Accept if $J_{\text{cand}} \leq J_{\text{cur}} - \varepsilon_{\text{acc}}$; otherwise reject and optionally backtrack. Maintain best solution z_{best} throughout.

E. Bellman Warm-Start and Local Re-solving

Control changes typically affect only local regions. We identify the affected region Ω^k_{chg} where $U_{s,r}(x;\tilde{z}) \neq U_{s,r}(x;z^k)$ or $\mathbf{M}_{s,r}(x;\tilde{z}) \neq \mathbf{M}_{s,r}(x;z^k)$. Using the old potential as initialization, iterate only within an expanded neighborhood $\Omega^{k,\delta}_{\text{chg}}$ until convergence.

This accelerates candidate evaluation. Periodic global corrections prevent error accumulation.

F. Algorithm Summary

1) Initialize potentials and density, compute $J(z^0 \mid \hat{p})$, set $z_{\text{best}} = z^0$.
2) For each iteration:
   - *Discrete update*: Generate neighborhood, screen with surrogate, evaluate top candidates with Bellman warm-start, select $s^{k+1/2}$.
   - *Continuous update*: Generate perturbations, compute stochastic gradient, generate candidate, accept/reject to obtain η^{k+1}.
   - *State update*: Set $z^{k+1} = (s^{k+1/2}, \eta^{k+1})$, compute objective, update z_{best} if improved.
3) Return $z^\star = z_{\text{best}}$.

Stopping criteria include: small objective change, no discrete variable change for consecutive iterations, or maximum iteration/simulation/time limits.

VII. Numerical Experiments

This section presents numerical experiments to validate the model mechanisms, evaluate control strategies, and demonstrate the optimization framework. The experimental design follows a two-layer validation logic: first verifying whether the model mechanisms authentically alter local crowd behaviors, then comparing different channel control strategies in terms of efficiency, safety, and load balancing.

A. Experimental Setup

1) Three-Channel Scenario: The experiments employ a three-channel scenario as the primary testbed for mechanism validation and strategy comparison. The geometry consists of an initial crowd region on the left, an exit region on the right, and three passages (upper, middle, lower) through a central wall separating the two regions.

Numerical parameters are set as follows: grid resolution 96×72, spatial step $\Delta x = 0.5$, free-flow speed $v_{\max} = 1.5$, maximum density $\rho_{\max} = 5.0$, initial density $\rho_{\text{init}} = 2.2$, CFL coefficient 0.9, maximum simulation steps 600, and time horizon $T = 40$.

2) Evaluation Metrics: The experiments employ three primary raw metrics and their normalized counterparts:

- *Efficiency metric* J_1 and $\tilde{J}_1$: total travel time and its mass-time normalized form;
- *Safety metric* J_2 and $\tilde{J}_2$: high-density exposure time and its space-time normalized form;
- *Balance metric* J_3 and $\tilde{J}_3$: channel flow variance and its normalized imbalance index.

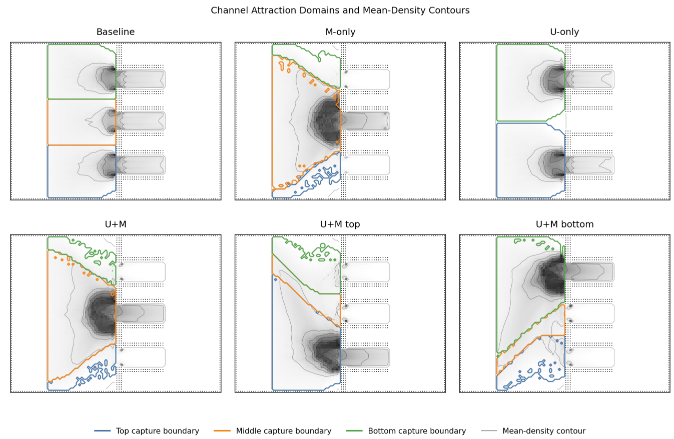

Fig. 2. Channel attraction domains and mean-density contours in the G1 experiment. Blue, orange, and green curves mark the capture-domain boundaries of the upper, middle, and lower channels, respectively. The grayscale background and thin gray lines show the time-averaged density field and its contours.

Behavioral indicators include channel capture domain ratios, flow distribution ratios, direction consistency indices, and approach angle distributions.

B. Mechanism Validation

1) Validation Objectives and Logic: The mechanism validation experiments address two fundamental questions. First, do geometric guidance $M(x)$ and directional constraints $U(x)$ authentically alter local decision-making and flow patterns of the crowd? Second, does the model produce sufficiently sensitive and interpretable behavioral responses when channel configurations change? Answering these questions is a prerequisite for subsequent strategy evaluation and optimization studies, since any optimization result built upon a model with distorted or unresponsive mechanisms would lack practical reliability.

The validation is organized into two parts. The first part verifies mechanism authenticity by comparing four configuration types—baseline, U-only, M-only, and combined $U + M$—to examine whether each mechanism induces the expected local behavioral changes. The second part verifies managerial applicability by testing whether the combined $U + M$ model yields clearly distinguishable behavioral responses to different channel control configurations (upper-channel-controlled, middle-channel-controlled, lower-channel-controlled).

2) Effect of Geometric Guidance $M(x)$: To verify whether geometric guidance alters channel approach patterns and attraction domain structures, we compare the baseline configuration (neither M nor U) against the M-only configuration, in which anisotropic guidance is applied to the middle channel. The remaining two channels are left uncontrolled.

Fig. 2 presents the channel attraction domains and mean-density contours across all G1 mechanism configurations. Under the baseline configuration, the three channels exhibit approximately balanced capture domains: upper 33.73%, middle 30.00%, lower 34.98%. Flow distributions are similarly balanced: upper 33.82%, middle 32.35%, lower 33.83%.

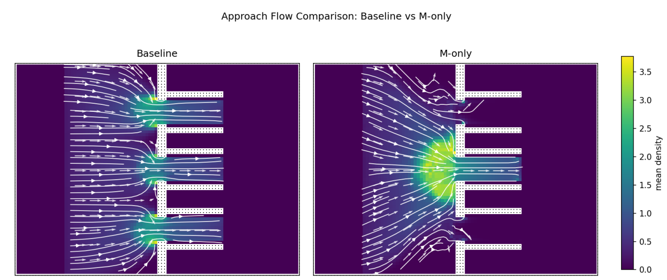

Fig. 3. Approach flow comparison between baseline (left) and M-only (right) configurations. White streamlines and arrows represent time-averaged movement directions; the background color encodes time-averaged density.

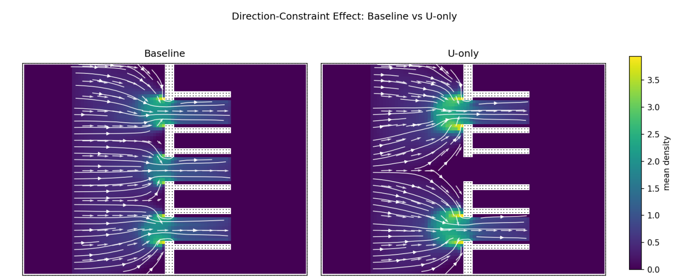

Fig. 4. Local direction field comparison between baseline (left) and U-only (right) configurations. White streamlines and arrows represent time-averaged movement directions; the background color encodes time-averaged density.

Under the M-only configuration, the middle channel capture domain expands significantly to 63.41%, and its flow share jumps to 99.32%. This demonstrates that geometric guidance $M(x)$ alone is sufficient to reshape local attraction domain structures, causing the crowd to converge toward the guided channel earlier and fundamentally altering the approach pattern.

The mechanism differences are directly visible in Fig. 2. Under the baseline configuration, the capture-domain boundaries of the three channels partition the feeder region roughly according to the channels' vertical positions, and mean-density hotspots appear near the entrances of all three channels, indicating that without guidance, the crowd naturally distributes itself based primarily on geometric distance. After introducing $M(x)$, the orange capture domain of the middle channel visibly expands upward and downward, and more concentrated high-density contours form in front of the middle channel entrance. This indicates that the anisotropic metric does not merely alter global indicators but reconstructs local attraction domains and approach paths ahead of the channel entrance. In contrast, the U-only panel mainly reflects the exclusion of channel accessibility by feasible-direction constraints, while the $U + M$ panels and their top- and bottom-channel variants show that capture domains migrate systematically with the position of the guided channel. The conclusion supported by Fig. 2 is therefore that the core behavioral effect of $M(x)$ is to alter *which channel attracts the crowd and when alignment with that channel begins*, rather than to explicitly forbid any path. This soft guidance mechanism provides a continuous and interpretable basis for flow reorganization under combined $U + M$ control.

Fig. 3 further reveals the effect of $M(x)$ on the approach process. Under the baseline configuration, the incoming flow from the left diverts toward the three channels only when approaching the central wall, exhibiting an overall pattern of near-parallel entry with localized turning at the channel openings. Under the M-only configuration, however, streamlines begin converging toward the middle channel at a noticeably earlier upstream position, forming a fan-shaped gathering pattern in front of the entrance and producing a more concentrated mean-density hotspot near the middle channel entrance. This confirms that geometric guidance alters the approach path and the turning position: pedestrians complete pre-alignment before entering the bottleneck, rather than merely changing the speed direction inside the channel.

TABLE I
BIDIRECTIONAL $U(x)$ VALIDATION RESULTS

Configuration	Westbound Share	Counter-flow Mass	Head-on Exposure
Bidirectional baseline	72.10%	188.15	887.70
Middle channel one-way	0	0	0

3) Effect of Directional Constraints $U(x)$: To verify whether directional constraints alter local optimal directions and channel selection structure, we compare the baseline configuration against the U-only configuration, in which the middle channel is restricted to eastward-only flow.

Under the U-only configuration, the middle channel capture domain drops to 0, and its flow share drops to 0, with the upper and lower channels splitting the traffic approximately $50/50$. This demonstrates that $U(x)$ significantly rewrites feasible path choices, but the unidirectional inflow experiment alone is insufficient to establish the mechanism's effectiveness in suppressing counter-flow and head-on conflicts.

Fig. 4 further illustrates this point. Under the baseline configuration, the direction field in front of the entrances points toward all three channels, and the middle channel maintains clear eastbound flow streamlines. Under the U-only configuration, the middle channel is excluded from the feasible choice set by the directional constraint, and the streamlines that would have approached the middle channel separate ahead of the entrance, diverting to the upper and lower passable channels. This figure demonstrates that the primary role of $U(x)$ is not smooth induction but rather the direct restructuring of reachable channel and path selection by modifying the local admissible direction set.

To further verify the suppression of counter-flow and head-on conflicts, we conduct a bidirectional flow sub-experiment. Two configurations are tested: a bidirectional baseline (westbound and eastbound crowds present simultaneously) and a bidirectional with middle channel one-way rule (middle channel restricted to eastward only).

The results in Table I show that when the middle channel enforces a one-way rule, westbound crowds are completely excluded from the middle channel, and the westbound share, counter-flow mass time, and head-on cell exposure within that channel are all driven to zero. This confirms that $U(x)$ not only

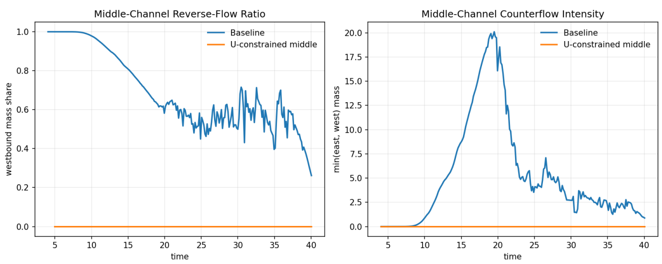

Fig. 5. Dynamic comparison of reverse flow in the middle channel under bidirectional conditions. Left: westbound mass share in the middle channel over time. Right: head-on conflict intensity in the middle channel over time.

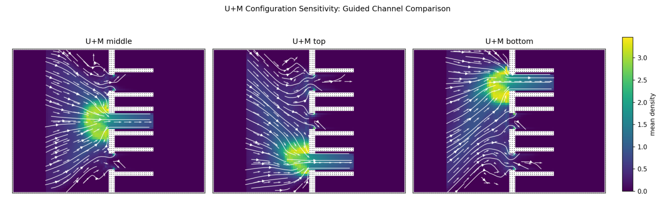

Fig. 6. Flow field comparison for $U+M$ combined mechanism under different channel control configurations. The three sub-figures correspond to middle-, upper-, and lower-channel-controlled setups, respectively. White streamlines and arrows represent time-averaged movement directions; the background color encodes time-averaged density.

TABLE II
COMBINED $U+M$ CONFIGURATION SENSITIVITY: CAPTURE DOMAINS AND FLOW SHARES

Configuration	Dominant Capture	Dominant Flow	Upper Capture	Middle Capture	Lower Capture	Upper Flow	Middle Flow	Lower Flow
$U+M$-middle	middle	middle	11.29%	62.63%	12.95%	0.40%	98.87%	0.72%
$U+M$-top	top	top	57.37%	12.76%	22.26%	95.15%	0.74%	4.11%
$U+M$-bottom	bottom	bottom	20.55%	11.98%	58.06%	2.98%	0.70%	96.31%

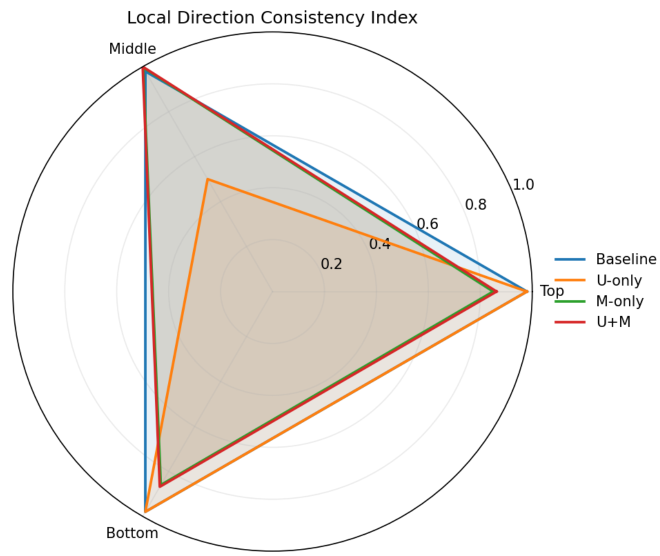

Fig. 7. Local direction consistency radar chart from the G1 experiment. The radar chart aggregates the local direction consistency index across the upper, middle, and lower channels under different mechanism configurations. Values closer to 1 indicate more concentrated local flow direction and stronger compliance with eastbound passage organization.

changes path selection but effectively suppresses counter-flow and head-on conflicts under bidirectional demand.

Fig. 5 presents the temporal dynamics. In the bidirectional baseline, the westbound share in the middle channel remains persistently high, accompanied by sustained head-on conflict mass. Once the middle channel one-way rule is imposed, both curves are rapidly driven to zero and remain stable throughout the observation window. This demonstrates that the suppression of bidirectional head-on conflicts by $U(x)$ is not a short-lived accidental outcome but a structural change that persists over the entire observation period.

4) Combined $U+M$ Effects and Configuration Sensitivity: To verify whether the combined $U+M$ model produces distinguishable behavioral responses to different channel configurations, we test three setups: middle-channel-controlled ($U+M$-middle), upper-channel-controlled ($U+M$-top), and lower-channel-controlled ($U+M$-bottom).

Table II reports the quantitative results. The $U+M$ model exhibits stable and interpretable responses to configuration changes: when the middle channel is controlled, it becomes the dominant capture channel (62.63%) and dominant flow channel (98.87%); when the upper channel is controlled, the upper channel becomes dominant (57.37% capture, 95.15% flow); and when the lower channel is controlled, the lower channel becomes dominant (58.06% capture, 96.31% flow). This demonstrates that the combined model is capable of distinguishing different control schemes and can serve as a reliable evaluation oracle for subsequent strategy comparison and optimization.

Fig. 6 shows that changing the controlled channel position induces synchronous migration of density hotspots and approach streamlines. When the middle channel is controlled, the crowd concentrates in front of the middle channel entrance and passes through it; when the upper channel is controlled, the dominant convergence region shifts to the upper channel entrance; and when the lower channel is controlled, it shifts to the lower channel entrance. These spatial patterns are consistent with the dominant capture and flow channels reported in Table II, confirming that the $U+M$ combined mechanism can not only express the local rules of a single channel but also produce distinguishable and interpretable global flow reorganization in response to different spatial configurations.

5) Mechanism Validation Conclusions: Fig. 7 supplements the preceding spatial flow-field findings from the perspective of directional organization. Under the baseline configuration, the three channels exhibit relatively similar direction consistency indices, reflecting that all channels carry a share of eastbound flow under natural distribution. The U-only configuration eliminates middle channel passage, so the middle channel consistency no longer forms a meaningful observation, while the upper and lower channels maintain high consistency. Under the M-only and $U+M$ configurations, the middle channel direction consistency reaches 0.992 and 1.000, respectively, indicating that when geometric guidance and directional constraints act jointly, not only is channel selection restructured, but the local directional organization within the channel also becomes more concentrated.

Synthesizing the experimental results from Fig. 2 through

Fig. 7 and Tables I–II, the G1 experiments support the following conclusions:

1) **Geometric guidance $M(x)$ significantly alters local approach patterns and channel attraction domains.** By reducing the effective travel cost along the channel axis, $M(x)$ causes the crowd to align with the channel direction earlier and converge smoothly, rather than making abrupt turns immediately before the entrance. This soft guidance mechanism provides a continuous, interpretable basis for flow reorganization.
2) **Directional constraints $U(x)$ reshape feasible path choices and directional organization.** $U(x)$ directly restructures the set of reachable channels and paths by modifying the local admissible direction set. Under bidirectional demand, it effectively suppresses counter-flow and head-on conflicts, with the suppression effect sustained throughout the entire observation period as a structural change rather than a transient outcome.
3) **The combined $U + M$ model exhibits clear sensitivity to different channel configurations.** It produces behavioral responses that are both rule-compliant and behaviorally realistic. Changing the controlled channel position induces synchronous migration of density hotspots, capture domains, and approach streamlines.
4) **The full model is necessary for reliable optimization.** If either U or M is removed, the model's response to configuration changes becomes either incomplete or behaviorally unrealistic. $U(x)$ answers which directions are allowed, while $M(x)$ answers which allowed directions better follow channel geometry. Subsequent optimization studies must therefore be built upon the joint mechanism.

C. Control Strategy Comparison

This subsection compares different channel control strategies in terms of normalized efficiency ($\tilde{J}_1$), normalized safety ($\tilde{J}_2$), and normalized load balancing ($\tilde{J}_3$). The strategies under comparison include:

- Baseline (unrestricted bidirectional channels);
- Six directional configurations, including single-entry and single-return strategy families;
- Three one-closed-channel configurations, where the upper, middle, or lower channel is temporarily closed and the remaining channels provide one entry and one return direction.

These cases test whether direction settings and closures produce distinguishable load redistribution, hotspot migration, and objective trade-offs. Detailed numerical results and analysis will be presented in the full version of this paper.

D. Multi-Stage Flow Validation Experiments

This subsection validates the multi-stage behavior chain and probabilistic splitting mechanism. The experiments compare single-stage approximations against full multi-stage models to demonstrate how stage transitions and fixed-probability splitting alter density evolution and exit load distribution.

Detailed results and analysis will be presented in the full version of this paper.

E. Optimal Control Parameter Search

This subsection presents the first-version automated parameter search results. The search enumerates discrete strategy variables, performs coarse grid search on guidance intensity η, and searches over fixed splitting probability candidates to identify optimal control configurations under different objective weight combinations $(\lambda_1, \lambda_2, \lambda_3)$.

Detailed results and analysis will be presented in the full version of this paper.

VIII. Conclusion

This paper presents a bi-level optimization framework for crowd channel management that combines discrete directional constraints with anisotropic geometric guidance. The proposed SA-HBO algorithm exploits problem structure through surrogate pre-screening, Bellman warm-start, and block coordinate updates to handle the expensive simulation-based optimization efficiently.

The numerical experiments validate that: (1) geometric guidance $M(x)$ and directional constraints $U(x)$ authentically alter local crowd behaviors; (2) the combined model exhibits stable sensitivity to configuration changes; (3) directional constraints effectively suppress counter-flow and head-on conflicts under bidirectional demand. These findings establish the foundation for subsequent strategy optimization studies.

Future work will extend to full multi-objective optimization, sensitivity analysis under demand perturbations, and calibration against real-world observation data.

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